

Ex. 2: Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, 1)$, with unknown μ .

a) If $n=1$, and we observe $X_1 = x_1$, then what is $\hat{\theta}_{MLE}(x_1)$?

$$\hat{\theta}_{MLE}(x_1) = \underset{\theta}{\operatorname{argmax}} f_x(x_1; \theta, 1) \quad \parallel \text{The pdf of a } N(\theta, 1) \text{ r.v.}$$

$$= \underset{\theta}{\operatorname{argmax}} \frac{1}{\sqrt{2\pi}} e^{-\frac{(x_1 - \theta)^2}{2}} \quad \left. \vphantom{\frac{1}{\sqrt{2\pi}}} \right\} \text{Drop the constants}$$

$$= \underset{\theta}{\operatorname{argmax}} e^{-\frac{(x_1 - \theta)^2}{2}} \quad \left. \vphantom{e^{-\frac{(x_1 - \theta)^2}{2}}} \right\} \text{Take log}$$

$$= \underset{\theta}{\operatorname{argmax}} -\frac{(x_1 - \theta)^2}{2} \quad \left. \vphantom{-\frac{(x_1 - \theta)^2}{2}}} \right\} \text{Drop constants}$$

$$= \underset{\theta}{\operatorname{argmax}} \underline{-(x_1 - \theta)^2}$$

$$\frac{d}{d\theta} = \underline{-2(x_1 - \theta)(-1)}$$

equals 0 at $\theta = x_1$

$$\boxed{\hat{\theta}_{MLE}(x_1) = x_1}$$

b) If $n > 1$ and we observe realizations $x_1, \dots, x_n$, what is $\hat{\theta}_{MLE}(x_1, \dots, x_n)$?

$$\hat{\theta}_{MLE}(x_1, \dots, x_n) = \underset{\theta}{\operatorname{argmax}} \prod_{i=1}^n \frac{1}{\sqrt{2\pi}} e^{-\frac{(x_i - \theta)^2}{2}}$$

$$= \underset{\theta}{\operatorname{argmax}} (2\pi)^{-n/2} \cdot e^{-\frac{1}{2} \sum_{i=1}^n (x_i - \theta)^2}$$

$$= \underset{\theta}{\operatorname{argmax}} \underline{-\sum_{i=1}^n (x_i - \theta)^2}$$

$$\frac{d}{d\theta} = \underline{-2 \sum_{i=1}^n (x_i - \theta)(-1)}$$

equals 0 at $\theta = \frac{1}{n} \sum_{i=1}^n x_i$

$$\boxed{\hat{\theta}_{MLE}(x_1, \dots, x_n) = \frac{1}{n} \sum_{i=1}^n x_i}$$

← MLE for unknown mean of a normally distributed pop. is the sample mean!

Ex. 3: Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Exp}(\lambda)$, with unknown λ .

For realizations $x_1, \dots, x_n$, find the MLE of λ .

$$\hat{\lambda}_{MLE}(x_1, \dots, x_n) = \arg\max_{\lambda} f(x_1; \lambda) \cdot \dots \cdot f(x_n; \lambda)$$

$$= \arg\max_{\lambda} \lambda \cdot e^{-\lambda x_1} \cdot \dots \cdot \lambda \cdot e^{-\lambda x_n}$$

$$= \arg\max_{\lambda} \lambda^n e^{-\lambda \sum_{i=1}^n x_i}$$

$$= \arg\max_{\lambda} \underbrace{n \cdot \log(\lambda) + (-\lambda \cdot \sum_{i=1}^n x_i)}$$

By defn of pdf for an $\text{Exp}(\lambda)$ r.v.

Exponent Rules

Take log & simplify

$$\frac{d}{d\lambda} = \frac{n}{\lambda} - \sum_{i=1}^n x_i$$

equals 0 when $\lambda = \frac{1}{\frac{1}{n} \sum_{i=1}^n x_i}$

$$\boxed{\hat{\lambda}_{MLE}(x_1, \dots, x_n) = \frac{1}{\frac{1}{n} \sum_{i=1}^n x_i}}$$

Notice: $E[\hat{\lambda}_{MLE}(X_1, \dots, X_n)] = E\left[\frac{1}{\bar{X}_n}\right] \neq \frac{1}{E[\bar{X}_n]} = \frac{1}{1/\lambda} = \lambda$

$\leadsto \hat{\lambda}_{MLE}(X_1, \dots, X_n)$ is biased!

So, if we observed:

$$x_1 = 2, \quad x_2 = 1/4, \quad x_3 = 1/2, \quad x_4 = 3/4$$

Then the MLE for λ is

$$\hat{\lambda}_{MLE}(2, 1/4, 1/2, 3/4) = \frac{1}{\frac{1}{4}(8)} = \underline{\underline{\frac{1}{2}}}$$

Confidence Sets:

The second unit of Inference!

In point estimation, we obtained a single "guess" for the value of an unknown population parameter θ . Our "guess" was a random point estimator $\hat{\theta}(X_1, \dots, X_n)$. We asked:

- What is $E[\hat{\theta}(X_1, \dots, X_n)]$? Is this estimator biased?
- What is $\text{Var}(\hat{\theta}(X_1, \dots, X_n))$? Is this small?

→ These mattered to us because $\hat{\theta}(X_1, \dots, X_n)$ is random!

+ $E[\hat{\theta}(X_1, \dots, X_n)] = \theta$ gives an estimator located around the true value

+ Small $\text{Var}(\hat{\theta}(X_1, \dots, X_n))$ means the estimator is usually close to its mean.

But we have two big problems:

1) It will (pretty much) always be the case that $\hat{\theta}(X_1, \dots, X_n) \neq \theta$.

2) Outside of $\text{MSE}(\hat{\theta}, \theta)$, we have no way of knowing how far the true θ is from $\hat{\theta}(X_1, \dots, X_n)$.

→ How can we resolve this??

An Idea: Intervals!

Instead of guessing a single (random) value of $\hat{\theta}(X_1, \dots, X_n)$, guess a random interval $[\underline{\theta}, \bar{\theta}]$ that is "very" likely to contain the true value of θ !

→ We want to find a way to choose $[\underline{\theta}, \bar{\theta}]$ for which

$IP(\theta \in [\underline{\theta}, \bar{\theta}]) = 1 - \alpha$ for some small $\alpha \in (0, 1)$ we specify.

Key Idea:

The interval $[\underline{\theta}, \bar{\theta}]$ is random! It should depend on the sample $X_1, \dots, X_n$. I.e. we have $[\underline{\theta}(X_1, \dots, X_n), \bar{\theta}(X_1, \dots, X_n)]$. In contrast, the parameter θ is not random!

Goal: Given α , determine $\underline{\theta}(X_1, \dots, X_n)$ and $\bar{\theta}(X_1, \dots, X_n)$ such that $IP(\underline{\theta}(X_1, \dots, X_n) \leq \theta \leq \bar{\theta}(X_1, \dots, X_n)) = 1 - \alpha$.

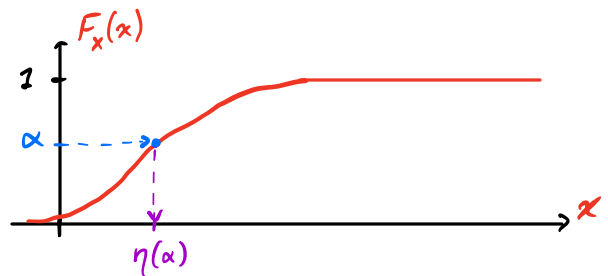
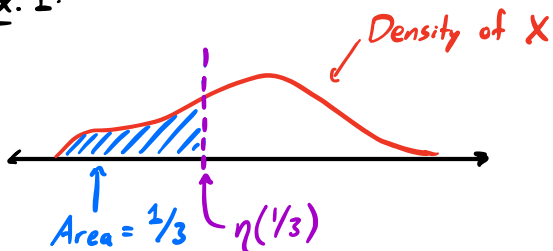
Preliminaries:

Def. (Percentile): For any $\alpha \in (0, 1)$, the $(100 \times \alpha)^{\text{th}}$ percentile of the a continuous random variable, denoted $\eta(\alpha)$, must satisfy:

$$F_X(\eta(\alpha)) = IP(X \leq \eta(\alpha)) = \alpha,$$

where $F_X(\cdot)$ is the CDF of X .

Ex. 1:



Ex. 2: Exponentially Distributed R.V.

If $X \sim \text{Exp}(\lambda)$, then $F_X(x; \lambda) = 1 - e^{-\lambda x}$

Using the def'n of percentile:

$$\alpha = F_X(\eta(\alpha); \lambda) = 1 - e^{-\lambda \eta(\alpha)}$$

$$\Rightarrow \eta(\alpha) = \underline{\underline{\frac{-1}{\lambda} \log(1 - \alpha)}}$$

Ex. 3: Standard Normal R.V.

If $X \sim N(0,1)$, then $F_X(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt$

Again:

$$\alpha = F_X(\eta(\alpha)) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\eta(\alpha)} e^{-t^2/2} dt$$

$$\leadsto \eta(\alpha) = ???$$

$\hookrightarrow$ We don't have an exact expression, so we must use software or tables!

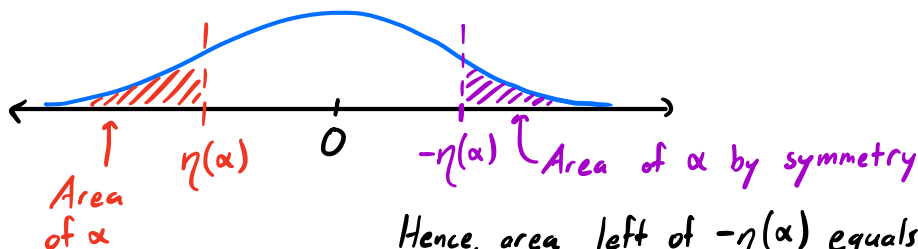
Common percentiles of standard normal r.v.'s:

α	0.025	0.05	0.1	0.5	0.9	0.95	0.975
$\eta(\alpha)$	-1.96	-1.65	-1.28	0	1.28	1.65	1.96

★ These will be very useful!

Note: For any normal r.v. with mean 0, we have $\eta(\alpha) = -\eta(1-\alpha)$.

In fact, this holds for any symmetric pdf around 0.



Hence, area left of $-\eta(\alpha)$ equals both $1-\alpha$, and $F(-\eta(\alpha))$

$$\leadsto \underline{1-\alpha} = F(-\eta(\alpha))$$

$F(\eta(1-\alpha))$ by def'n.

$$\Leftrightarrow F(-\eta(\alpha)) = F(\eta(1-\alpha))$$

$$\therefore \underline{-\eta(\alpha) = \eta(1-\alpha)}$$

Claim: Let $X \sim N(\mu, \sigma^2)$, and $Z = \frac{X-\mu}{\sigma}$, such that $Z \sim N(0,1)$.

If $\eta_X(\alpha)$ are percentiles of X , and $\eta_Z(\alpha)$ are percentiles of Z ,

then: $\eta_X(\alpha) = \mu + \sigma \cdot \eta_Z(\alpha)$ for all α .

Proof:

$$\alpha = P(X \leq \eta_X(\alpha)) \quad \text{by def'n of } \eta_X(\alpha)$$

$$= P(\mu + \sigma \cdot Z \leq \eta_X(\alpha)) \quad \text{by } Z = \frac{X - \mu}{\sigma}$$

$$\alpha = P\left(Z \leq \frac{\eta_X(\alpha) - \mu}{\sigma}\right) \quad \text{rearranging}$$

$$\therefore \underline{\eta_Z(\alpha) = \frac{\eta_X(\alpha) - \mu}{\sigma}} \quad \text{by def'n of } \eta_Z(\alpha)$$

The standard normal distribution is really special, so we give a slight variation of its percentiles a special name.

Def. (Critical Value): Let $Z \sim N(0,1)$ be a standard normal random variable. For any $\alpha \in (0,1)$, the α critical value z_α solves $\alpha = P(Z \geq z_\alpha)$.

Note: z_α is the $(100 \times (1-\alpha))^{\text{th}}$ percentile.

Observation:

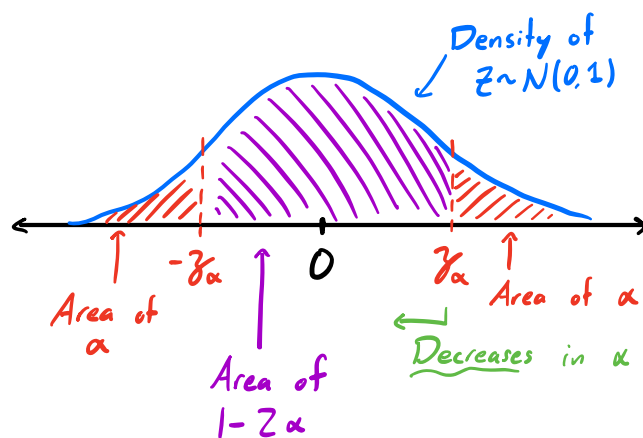
$$\bullet P(Z \leq z_\alpha) = 1 - \alpha$$

$$\bullet P(Z \geq -z_\alpha) = 1 - \alpha$$

$$\bullet P(-z_\alpha \leq Z \leq z_\alpha) = 1 - 2\alpha$$

More often, we write the third as:

$$P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1 - \alpha$$



Def. (Z-Score): Let x be a realization of a random variable X with mean μ and variance σ^2 . The z -score of x is equal to

$$z := \frac{x - \mu}{\sigma}, \quad \text{i.e. } z = \# \text{ of standard deviations (of } X) \text{ between } x \text{ and } \mu = E[X].$$

Confidence Intervals:

Def. (Confidence Interval): Let θ be an unknown parameter, and $X_1, \dots, X_n$ a random sample. For error level $\alpha \in (0, 1)$, we say that $[\underline{\theta}(X_1, \dots, X_n), \bar{\theta}(X_1, \dots, X_n)]$ is a $1-\alpha$ confidence interval for θ if:

$$P(\underline{\theta}(X_1, \dots, X_n) \leq \theta \leq \bar{\theta}(X_1, \dots, X_n)) = 1 - \alpha.$$

i.e., " θ lies outside $[\underline{\theta}, \bar{\theta}]$ with probability α "

Note: The interval $[\underline{\theta}(X_1, \dots, X_n), \bar{\theta}(X_1, \dots, X_n)]$ is random here, not the parameter θ . This differs from our usual statements about "P(a R.V. is between two constants)".

Note: Looks similar to percentiles / critical values, but here θ is not random. Still, they will help us!

Note: A tradeoff!

Smaller $\alpha \Leftrightarrow$ Wider Interval

Larger $\alpha \Leftrightarrow$ Narrower Interval

Common choices of α are 10%, 5%, 1%.

To build confidence intervals (LI's), we need to know something about distributions - specifically of $\underline{\theta}(X_1, \dots, X_n)$ and $\bar{\theta}(X_1, \dots, X_n)$.

$\leadsto$ The best tool we have here is the Central Limit Theorem.
